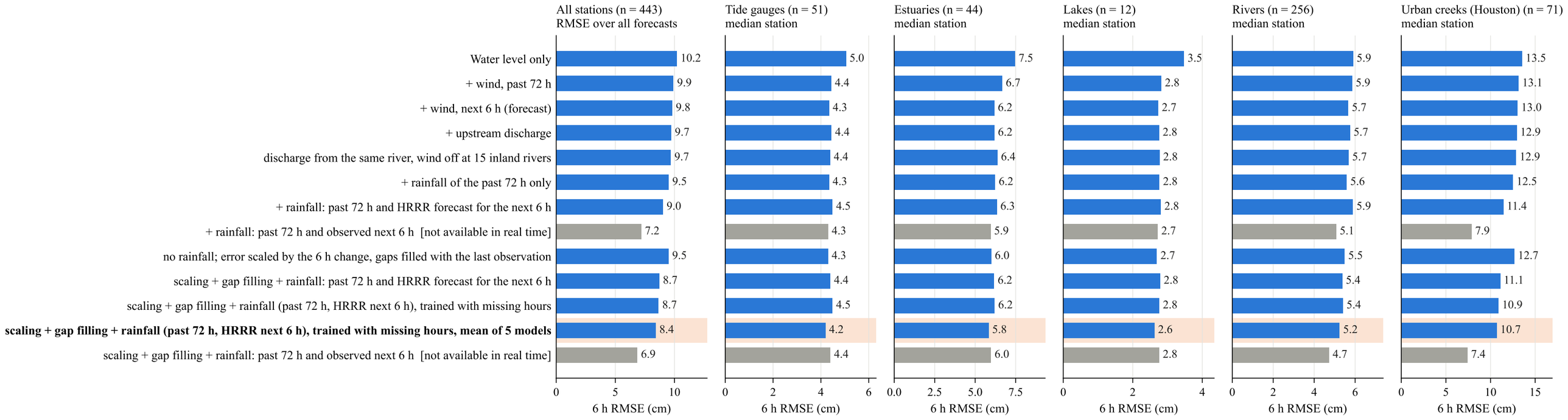
All stations (n = 443)
RMSE over all forecasts
Tide gauges (n = 51)
median station
Estuaries (n = 44)
median station
Lakes (n = 12)
median station
Rivers (n = 256)
median station
Urban creeks (Houston) (n = 71)
median station
Water level only
+ wind, past 72 h
+ wind, next 6 h (forecast)
+ upstream discharge
discharge from the same river, wind off at 15 inland rivers
+ rainfall of the past 72 h only
+ rainfall: past 72 h and HRRR forecast for the next 6 h
+ rainfall: past 72 h and observed next 6 h [not available in real time]
no rainfall; error scaled by the 6 h change, gaps filled with the last observation
scaling + gap filling + rainfall: past 72 h and HRRR forecast for the next 6 h
scaling + gap filling + rainfall (past 72 h, HRRR next 6 h), trained with missing hours
scaling + gap filling + rainfall: past 72 h, HRRR next 6 h, trained with missing hours, mean of 5 models
scaling + gap filling + rainfall: past 72 h and observed next 6 h [not available in real time]
10.2 9.9 9.8 9.7 9.7 9.5 9.0 7.2 9.5 8.7 8.7 8.4 6.9
5.0 4.4 4.3 4.4 4.4 4.3 4.5 4.3 4.3 4.4 4.5 4.2 4.4
7.5 6.7 6.2 6.2 6.4 6.2 6.3 5.9 6.0 6.2 6.2 5.8 6.0
3.5 2.8 2.7 2.8 2.8 2.8 2.8 2.7 2.7 2.8 2.8 2.6 2.8
5.9 5.9 5.7 5.7 5.7 5.6 5.9 5.1 5.5 5.4 5.4 5.2 4.7
13.5 13.1 13.0 12.9 12.9 12.5 11.4 7.9 12.7 11.1 10.9 10.7 7.4
0 5 10
0 2 4 6
0.0 2.5 5.0 7.5
0 2 4
0 2 4 6
0 5 10 15
6 h RMSE (cm)